

# RAHUL KUMAR

Software Engineer | Chandigarh, India

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## Summary

Software Engineer experienced shipping production systems in **TypeScript, Go, and Rust**. Skilled at crafting polished, accessible frontend components in React/Next.js, building reliable services in **Rust**, and engineering high-performance APIs and data pipelines end-to-end, with a focus on **caching, security, and low-latency query scaling**.

## Technical Skills

**Languages:** TypeScript, JavaScript, Go, Rust, SQL

**Frontend:** Next.js, React.js, Redux Toolkit, Tailwind CSS, Shadcn UI

**Backend:** Node.js, Express.js, Fastify, Echo, Fiber, gRPC, WebSockets, REST APIs

**Databases:** PostgreSQL, MongoDB, ClickHouse, Redis

**Infrastructure & DevOps:** Docker, GitHub Actions, AWS (SES), Prisma, Drizzle ORM, BullMQ

**Core Concepts:** Data Structures & Algorithms, System Design, Distributed Systems, Microservices

## Experience

### InboxKit

Jan 2026 – Present

*Full-Stack Software Engineer*

*Remote*

*Tech: Go, Fiber, Fastify, Next.js, ClickHouse, Redis, BullMQ, Puppeteer, AWS SES, Docker, Mongo DB*

- **Owned the Enrich Platform end-to-end:** designed and shipped the full product (frontend, backend, infra, deployment), including the REST API surface powering the customer dashboard and an internal admin dashboard for monitoring and abuse detection.
- Designed and built the customer-facing UI in Next.js, Tailwind, and Shadcn: dashboards, billing, analytics, and a reusable component library with accessible, responsive layouts across breakpoints.
- Architected ClickHouse-backed analytics over **300M+ records**, applying **multi-tier Redis caching**, query rewriting, and partitioning to keep dashboard queries at **sub-second p95 latency**.
- Hardened platform **security**: rate limiting, RBAC, signed sessions, strict input validation, and audit logging to prevent abuse, injection, and account takeover.
- Built microservices and automation pipelines (browser automation, transactional email, LinkedIn scraping) with Redis-backed BullMQ queues handling retries, backoff, and rate-limited dispatch.

## Key Projects

**DevSync** – Real-Time Collaborative Coding Platform    *Next.js, Prisma, Yjs, Xvfb, WebSocket, Express, Docker*    Live Demo

- Built a Next.js collaborative coding platform supporting up to **15 developers editing the same repository simultaneously** in real time via Yjs CRDTs over WebSocket, with low-latency cursor and selection sync.
- Architected a distributed microservices system separating collaboration (WebSocket, CRDT sync, presence) from persistence and execution layers for horizontal scalability.
- Developed a Dockerized execution backend with isolated disk storage, capable of running React, Express, and GUI apps (Tkinter), exposed securely via a reverse proxy.
- Integrated a real-time chat system alongside the editor for seamless team communication during sessions.

**GoRedis** – Redis-Compatible In-Memory Database (from Scratch)    *Go, TCP, RESP Protocol*

GitHub

- Built a Redis-compatible database engine in Go with a hand-written **RESP protocol parser** and a goroutine-per-connection server, implementing **40+ commands** so the official **redis-cli** and client SDKs connect natively at sub-millisecond GET/SET latency.
- Implemented core data structures from scratch: strings with TTL, lists with blocking ops (LPUSH/RPUSH/BLPOP), streams (XADD/XRANGE/XREAD), sorted sets via skip-lists (ZADD/ZRANGE/ZRANK), and geospatial indexing (GEOADD/GEORADIUS).
- Engineered **MULTI/EXEC transactions** with WATCH-based optimistic concurrency control, plus **master-replica replication** (PSYNC/WAIT), **RDB snapshots**, **AOF write-ahead logging**, Pub/Sub fan-out, and **ACL authentication**.

**BookWise** – Enterprise Library Management System    *Next.js, Upstash, Drizzle, Redis, Nodemailer*

Live Demo

- Implemented a protected admin dashboard for user approvals, inventory management, and borrowing workflows.
- Integrated QStash background workflows with Nodemailer to automatically send overdue-book reminder emails.
- Hardened the API surface with rate limiting and request throttling to mitigate abuse and DDoS risk.

## Education

**Bachelor of Computer Applications (BCA)**

*DAV, Chandigarh, India*

2024 – 2027